

CS5275 The Algorithm Designer's Toolkit

S2 AY2025/26

Tutorial problem set 3

(There is no need to submit your solution for tutorial problem sets)

Problem 1 (PCP theorem and hardness of approximating 3SAT). Show that the statement

$$\text{NP} = \text{PCP}(O(\log n), O(1))$$

implies the following hardness of approximation result:

There exists a constant $\varepsilon > 0$ such that it is NP-hard to distinguish between the following two types of 3SAT instances:

- the formula is satisfiable,
- at most a $(1 - \varepsilon)$ fraction of the clauses can be satisfied.

Problem 2 (Characterization via level-1 Fourier weight). Let $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ be a Boolean function. Define the level-1 Fourier weight

$$W^1[f] = \sum_{i=1}^n \widehat{f}(\{i\})^2.$$

Prove that if

$$W^1[f] = 1,$$

then f is a dictator function $f(x) = x_i$ or its negation $f(x) = -x_i$, for some $i \in [n]$.

Problem 3 (Properties of recursive majority). Let n be an **odd** integer. Define the *recursive majority function of depth d*

$$f : \{-1, 1\}^{n^d} \rightarrow \{-1, 1\}$$

as follows: Partition the input into n blocks, compute the recursive majority on each block of depth $d - 1$, and output the majority of these values.

Determine whether f satisfies each of the following properties:

1. Monotone
2. Odd (i.e., $f(-x) = -f(x)$)
3. Unanimous (i.e., $f(1, 1, \dots, 1) = 1$ and $f(-1, -1, \dots, -1) = -1$)
4. Symmetric

Problem 4 (Influence of the monotone chain function). Let $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ be defined by

$$f(x) = 1 \quad \text{if and only if} \quad x_1 \leq x_2 \leq \dots \leq x_n.$$

Compute the total influence $I[f]$ exactly as a function of n .